

PART A-

- **Title of Micro Project:**

Prepare case study on iot based Smart Stick for Blind person

1.0 Brief Introduction:

The IoT-based Smart Stick for blind individuals is a cost-effective and innovative solution designed to enhance mobility and safety. It integrates an ultrasonic sensor for obstacle detection, a flame sensor to identify fire hazards, and an ESP32 microcontroller for smart processing. A buzzer and vibration motor provide real-time alerts, ensuring the user's awareness of nearby dangers. Additionally, the Pushbullet API sends emergency notifications to caretakers. This project empowers visually impaired individuals with greater independence and improved situational awareness.

2.0 Aim of the Micro-Project :

- State IoT Issues and challenges in deployment.
- Describe designs of IoT.

3.0 Course outcome :

- Interpret Internet of Things concepts.
- Describe IoT systems in which information and knowledge are inferred from data.

4.0 Action Plan

Details of Activity	Planned Start Date	Planned finish date	ID.Code & Name of team member
1)Discussion of micro-project	15/1/2025	24/1/2025	All members
2) Planning of software development.	25/1/2025	19/2/2025	22CM010-Naman Bharsakale 22CM0016-Sarvesh Chaudhari
3)Activities such as analyzing and monitoring are performed.	20/2/2025	29/2/2025	22CM010-Naman Bharsakale 22CM0016-Sarvesh Chaudhari
4) Completion of microproject and report.	30/2/2025	10/3/2025	22CM010-Naman Bharsakale 22CM0016-Sarvesh Chaudhari
5)Finalizing and checking.	30/3/2025	25/4/2025	22CM010-Naman Bharsakale 22CM0016-Sarvesh Chaudhari

5.0 Team member Name & ID CODE

Team member Name	ID CODE	Signature
1) NAMAN BHARSAKALE	22CM010	
2) SARVESH CHAUDHARI	22CM016	

Guideline for Assessment of Micro-Project

Evaluation as per suggested Rubric for Assessment of Microproject

Assessment Parameter	Characteristic to be assessed	Average (1 mark)	Good (1.5 mark)	Excellent (2 mark)
Process Assessment (06)	Relevance of the courses & proposals			
	Literature survey/market survey/information collection			
	Analysis of data & completion of the target as per proposal/			
Product Assessment (04)	Report Preparation/Quality of Prototype/model			
	Presentation / Viva			

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PART B

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4.0 Resources required

Sr.No.	Name of resources	Specifications	Qty	Remarks
1)	Printer	For printing microproject report.	1	-
2)	Website - Google, Wikipedia	For collecting information & references.	1	-
3)	Software	Compilers ,android application, and other documented.	2	
4)	Computer & smartphone	To gathering the information about microproject .	2	

5.0 Outputs of the Micro-Projects

- **IoT Communication:** Understood how to use Pushbullet API for wireless communication and emergency alert systems.
- **Problem Solving:** Applied logical thinking to address real-world problems faced by visually impaired individuals.

6.0 Skills Developed/learning out of this Micro-Project

- **Embedded System Programming** – Writing and uploading code to ESP32 using Arduino IDE or MicroPython .
- **IoT Integration** – Implementing internet-based features using Pushbullet API for emergency alerts.

7.0 Signature of Student

Team member Name	ID CODE	Signature
1) NAMAN BHARSAKALE	22CM010	
2) SARVESH CHAUDHARI	22CM016	

8.0 Assessment by Faculty as per Rubrics

Process Assessment (06)	Product Assessment (04)	Total Marks (10)	Signature of Faculty